澳門理工大學
Universidade Politécnica de Macau
Macao Polytechnic University

應用科學學院
Faculdade de Ciências Aplicadas
Faculty of Applied Sciences

Doctor of Philosophy in Computer Applied Technology

Thesis Defence Examination

Internal Ballot Form

Student Name Na Han

Student ID P2314691

	Pass	Minor Revision	Major Revision	Fail
Vote				

Put an "X" in the cell to vote.

Item	***Subitem***	***Mark ****			
		Excellent	**Good**	**Fair**	**Poor**
Thesis	*Knowledge Base and Literature Review* [1]				
	Originality and Independence [2]				
	Arguments and Reasoning [3]				
	Evaluation within the Field [4]				
Oral Defence	*Presentation* [5]				
	Discussion [5]				
Both	*Completeness and Coherence* [4,5]				

*Put an "X" in the cell to mark. 1-5: See the remarks on Sheet2. * See the marking guidelines on Sheet2.*

Additional Comment

Examiner Prof. Yanwei Chen

Date 09 July 2026

Panel Member Type	***Member Name***	***Date***
Chair	Prof. Lin Chen	9/7/2026
Examiner	Prof. Yanwei Chen	
Supervisor	Dr. Yapeng Wang	
	Dr. Hong Lin	
	Qiu Huang	

Remarks

(Marking on the subitems is based on MACAO POLYTECHNIC UNIVERSITY DOCTORAL THESIS EXAMINATION GUIDELINES)

Article 3: CRITERIA FOR AWARD

The thesis and the defence together should demonstrate:

1. a deep and thorough understanding of knowledge of the relevant discipline and the context within which the research is grounded and of the literature relevant to the research;
2. the originality and independence of the candidate’s thought, approach, investigation, analysis and result;
3. the sources from which information has been derived;
4. the exercise of critical judgment with regard to both the candidate's own work and that of other scholars in the field, and
5. that it is an integrated whole and presents a coherent argument in an appropriate form, both orally and in writing.

Marking Guidelines

	Excellent	Good	Fair	Poor
Knowledge Base and Literature Review	A comprehensive picture of the field is shown, with deep insights into the current progresses and challenges. A thorough review of the literature is given, that clearly uncovers the significance, applications, approaches and inspirations leading to the topics of the thesis.	The field is essentially introduced and illustrated. The current progresses and challenges are clearly stated and analyzed. The literature is effectively reviewed, revealing the significance, applications, and approaches, related to the topic of the thesis.	The field is briefly introduced, slightly out of focus. The current progresses and challenges are acceptably stated. The analysis is lacking and inaccurate. The literature is reviewed to bring up the significance, applications and approaches, showing some links to the topics of the thesis.	The field is poorly introduced. The current progresses and challenges are not clear. The literature review does not help in showing the significance, applications, and approaches, with weak links to the topic of the thesis.

Originality and Independence	The innovation and contribution are clearly identified and summarized, with justified significance. Evidences of originality are sufficient and well documented, including publications, patents, and/or copyright registrations. Original technical details are clearly formalized and explained, including propositions, equations, algorithms, and/or architectures.	The innovation and contribution are stated, and the significance is discussed. Evidences of originality are provided and linked to the topic, including publications, patents, and/or copyright registrations. Original technical details are well formulated and explained, including propositions, equations, algorithms, and/or architectures.	The innovation and contribution are listed, however the significance is not well explained. Evidences of originality are merely provided, including publications, patents, and/or copyright registrations. Original technical details are described in length, however lacking some rigor and appropriate forms, in propositions, equations, algorithms, and/or architectures.	The innovation and contribution are not clear. Evidences of originality are not provided or severely insufficient or irrelevant, in publications, patents, and/or copyright registrations. Original technical details are omitted or only provided with inappropriate forms, in propositions, equations, algorithms, and/or architectures.
Arguments and Reasoning	The argumentation is complete and well structured, including clear documentation of proofs, datasets, experiments, test suites, and result analysis. The conclusion is appropriate and accurate. Theoretical justification of the soundness and generality is provided.	The argumentation is thorough and organized, covering essential sources to draw conclusions, including proofs, datasets, experiments, test suites, and result analysis. The conclusion is appropriate. Analytical discussion of the soundness and generality is provided.	The argumentation is provided in detail, however some weakness can be seen in the reasoning and/or arguments. Conclusions are justified, with acceptable sources, including proofs, datasets, experiments, test suites, and result analysis. Discussion of the soundness and generality is merely provided.	The argumentation is incomplete and unacceptable, the reasoning and arguments are insufficient. The conclusion is not appropriate, without proper proofs, datasets, experiments, test suites, and/or result analysis. Discussion of the soundness and generality is omitted.
Evaluation within the Field	Quantified application-based performance benchmarking is conducted. Comparisons with the baseline and state-of-the-art are provided. Strengths and weaknesses are analyzed with complete justification. Outlooks into the future of the field are well discussed.	Quantified application-based performance benchmarking is conducted. Comparisons with the baseline and state-of-the-art are provided. Strengths and weaknesses are discussed in detail. Future research directions in the field are well projected.	Performance benchmarking is conducted. Comparisons with the baseline and/or state-of-the-art are merely given. Strengths and weaknesses are discussed, however insufficient. Future research directions in the field are vaguely mentioned.	Performance benchmarking is not conducted. Comparisons with the baseline and/or state-of-the-art are not appropriately provided. Strengths and weaknesses are unclear. Future research directions in the field are omitted.

Presentation	The presentation is attractive and well organized with rhythm and passion, highlighting the major challenges and contribution. The most difficult and brilliant points are introduced with insights, in a way that is interesting and easy to follow. The demonstration is stylish, aesthetic, elegant and pleasing.	The presentation is smooth and organized, highlighting the major challenges and contribution. All key points are well introduced with insights, and easy to understand. The demonstration is complete, well designed, rigorous, and concise.	The presentation is acceptable and structured, covering the major challenges and contribution. Key points are introduced and explained. The demonstration is consistently integrated and relevant, however the description contains apparent weakness.	The presentation is not acceptable and ill-structured, lacking the major challenges and contribution. Key points are not introduced or explained. The demonstration is inconsistent and irrelevant. The description is flawed.
Discussion	The discussion shows confidence and relaxation. Answers and explanations are concise and straightforward. Insightful details and breadth of knowledge are demonstrated. Future research momentum and determination are well displayed.	The discussion shows confidence and determination. Answers and explanations are flawless and straightforward. Details of thoughts and essential knowledge are demonstrated. Future research momentum and interests are exhibited.	The discussion shows some confidence. Answers and explanations are acceptable, some require clarification. Thoughts and knowledge are demonstrated, however with some weakness. Future research interests are mentioned and relevant.	The discussion shows no confidence and certainty. Answers and explanations are missing, insufficient or unacceptable. Thoughts and knowledge are not demonstrated or contain obvious flaws. Future research interests and motivation are omitted.
Completeness and Coherence	The thesis is complete and well integrated, with each part being relevant, telling a whole story. The oral presentation is coherent to the thesis, giving a vivid introduction and promotion, that helps in the mastery, and inspires the discussion, of the main topic.	The thesis is complete and structured, showing clear connections between the components, all belonging to one story. The oral presentation is coherent to the thesis, effectively introduces and promotes the ideas of the main topic, leading well into the discussion.	The thesis is mostly comprehensive, showing explicit connections between the components, with an apparent focus. The oral presentation is related to the thesis, discusses the ideas of the main topic, making the points in the discussion. It is also detectable that the work is cobbled together in a degree.	The thesis is not an integrated whole, showing no clear connections between the components, with no apparent focus. The oral presentation diverges from the thesis, discussing off-topic items, making confusion in the discussion. It is obvious that the work is completely cobbled together.